

AI-Powered Stock Portfolio Optimiser

by

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Data Science Project: Software Development with Big Data and Cloud

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Contents

0.1 Abstract 2

0.2 Introduction 3

0.3 Literature Review 4

0.4 Research Methodology 5

0.5 System Design and Architecture 6

0.6 Implementation 7

0.7 Data Analysis and Results 8

0.8 Discussion 9

0.9 Conclusion 11

0.1 Abstract

This project presents the development of an AI-powered stock portfolio optimiser that integrates Data Science, Big Data techniques, and Cloud Computing. The goal was to create an intelligent tool that could analyse high-frequency stock data, optimise portfolio allocations, and use Value at Risk (VaR) calculations to manage risk. Data cleansing, Monte Carlo simulations, Sharpe Ratio maximisation, and cloud deployment using Streamlit were all part of the technique. An interactive web dashboard was used to display the results of the simulation and data preparation. Important results show that such a system is useful for financial decision support, scalable, and feasible ([Glasserman 2004](#), [Markowitz 1952](#), [Sharpe 1966](#)). The research methodology, implementation difficulties, and prospects for the future are all covered in detail in this report.

0.2 Introduction

Big Data and Cloud Computing are transforming industries worldwide by providing scalable, efficient, and cost-effective solutions for data storage, analysis, and management (Mell & Grance 2011). Businesses and organisations create enormous volumes of data in the connected world of today, and in order to derive valuable insights, reliable technologies are needed. Cloud computing is a crucial platform for big data applications because it provides on-demand access to processing power and storage resources (Armbrust et al. 2010). In the banking industry, where real-time data analysis facilitates crucial decision-making procedures like risk assessment and portfolio management, the applicability of these technologies is particularly clear (McAfee & Brynjolfsson 2012).

The research problem addressed in this project is the need for an intelligent, automated tool that can optimise stock portfolios by leveraging data-driven methods. Traditional manual approaches to portfolio construction are often time-consuming, subjective, and prone to human error (Fabozzi et al. 2007). Investors must also balance risk and return in an environment where markets are volatile and unpredictable. Therefore, there is a clear demand for a cloud-based, scalable solution that applies quantitative finance principles to assist users in building optimised investment portfolios.

The main objectives of this project are to develop a stock portfolio optimiser using Modern Portfolio Theory (MPT) (Markowitz 1952), apply Monte Carlo simulations to explore thousands of potential portfolio configurations (Glasserman 2004), incorporate Value at Risk (VaR) calculations to assess potential losses (Jorion 2007), and deploy the entire system on a cloud platform (Streamlit Cloud) for easy access and usability. The project also uses GitHub to maintain version control and modular code in order to guarantee scalability, transparency, and repeatability.

The scope of this research is focused specifically on the intersection of Big Data analytics and Cloud Computing within the domain of financial portfolio optimisation. It involves analysing sizable historical datasets of minute-by-minute trade data from top tech firms, creating optimal portfolios using statistical and machine learning methods, and implementing the solution through an interactive dashboard hosted in the cloud. Although this project does not go into predictive analytics or real-time data streaming, it does set the stage for future developments.

This study is important because it has the potential to use cloud platforms and big data approaches to democratise access to smart investment tools. The project demonstrates the useful real-world uses of Big Data and Cloud Computing by developing a solution that is both technically strong and openly available. It illustrates how people and smaller organisations can use sophisticated financial analytics without requiring a lot of infrastructure or technical know-how, which promotes financial empowerment and increased financial literacy.

0.3 Literature Review

One of the fundamental ideas in financial economics, Modern Portfolio Theory (MPT) was first presented by Harry Markowitz in 1952 and focusses on optimising a portfolio's expected return given a specific amount of risk ([Markowitz 1952](#)). By mathematically proving that combining assets with different levels of correlation can lower total portfolio risk without necessarily compromising returns, Markowitz's approach popularised the idea of diversification. The portfolio optimisation strategies used in this project are supported by this theory, which is still a fundamental component of portfolio management today ([Elton et al. 2011](#)).

William F. Sharpe established the Sharpe Ratio in 1966 as a common tool for assessing investment portfolio performance on a risk-adjusted basis, further expanding on risk-return frameworks ([Sharpe 1966](#)). By calculating the excess return per unit of risk, the Sharpe Ratio enables investors to compare assets or portfolios with different levels of volatility in a consistent manner. In quantitative finance and investment analysis, the Sharpe Ratio and MPT are still frequently used today ([Bodie et al. 2014](#)).

Glasserman (2004) popularised Monte Carlo simulation techniques in financial engineering, which provide strong instruments for assessing portfolio performance in a variety of random market conditions ([Glasserman 2004](#)). Monte Carlo methods improve the robustness of portfolio decision-making processes by simulating thousands of possible future outcomes based on historical return distributions. This allows for insights into likely returns, risks, and potential severe losses.

One of the most widely utilised risk measures in institutional risk management procedures as well as regulatory frameworks is Value at Risk (VaR). VaR offers a measurable method of evaluating portfolio risk by estimating the greatest expected loss over a specific time period at a predetermined confidence level ([Jorion 2007](#)). Financial institutions have embraced it in large numbers in response to Basel II and Basel III regulations ([Basel Committee on Banking Supervision 2011](#)).

In terms of technological advancements, Big Data tools such as Pandas, NumPy, and SciPy have enabled efficient processing and analysis of massive datasets ([McKinney 2012](#)). An open-source Python library called Streamlit has become a very useful tool for quickly deploying dashboards, enabling developers to create dynamic data apps with no overhead ([Streamlit Inc. 2022](#)). Big data applications can be hosted in scalable environments on cloud platforms like Streamlit Cloud, Amazon Web Services (AWS), and Google Cloud Platform (GCP), which give users access without requiring local infrastructure ([Google Cloud 2024](#)). GitHub makes version control and collaborative development easier, which is crucial for contemporary cloud-based applications ([Chacon & Straub 2022](#)).

Although there are strong frameworks for risk management and portfolio optimisation in the literature currently in publication, and technological developments have made big data processing and cloud deployment more accessible, there is still a need to fully integrate these components into an intuitive, open-access educational tool. The majority of MPT and Monte Carlo simulations' academic uses are still theoretical or limited to institutional settings with intricate user interfaces ([Demanyk & Kolliner 2018](#)). By fusing big data processing, cloud deployment, and financial theories, this project seeks to close that gap and provide a simplified, approachable

tool that can be used for both teaching and raising general financial awareness.

0.4 Research Methodology

This project adopts an experimental research design, focusing on the data-driven analysis of financial markets to construct optimised investment portfolios. The approach involves systematically applying statistical simulations and risk modelling techniques to historical stock market data, testing thousands of potential portfolio configurations, and evaluating their performance based on quantitative metrics (Baxter & Rennie 2001). The methodology is experimental in nature because it uses real-world data to simulate different investment scenarios and observe their outcomes under controlled, repeatable conditions.

The project makes use of Streamlit Cloud for the deployment of the cloud platform component. Without needing consumers to set up complicated server infrastructure, Streamlit Cloud offers a smooth environment for hosting Python-based data science applications, making the optimiser available everywhere (Streamlit Inc. 2022). GitHub is used for collaborative project management and version control (Chacon & Straub 2022). Despite consideration, industry-standard big data solutions such as Apache Hadoop or Spark were judged unnecessary for processing data at this scale. Instead, Python modules like Pandas and NumPy, which are ideal for structured financial datasets of this size, were used to handle data manipulation efficiently (McKinney 2012).

Data collection involved sourcing minute-level trading datasets in CSV format for four major technology companies: Apple (AAPL), Amazon (AMZN), Google (GOOGL), and Tesla (TSLA). Microsoft (MSFT) was originally included but excluded later due to insufficient recent data coverage. These datasets, covering multiple years of trading activity, were processed locally and uploaded to the project repository. Each dataset included key fields such as Date, Time, Open, High, Low, Close, and Volume, which were essential for reconstructing stock price movements and calculating returns.

Preprocessing the data to merge the Date and Time columns into a single Datetime index, filtering records to keep only data from the previous five years, and removing any inconsistent or missing items made up the experiment setup (Tsay 2010). Then, thousands of possible portfolio weight combinations were generated at random to perform Monte Carlo simulations, and the predicted return and volatility for each simulated portfolio were determined (Glasserman 2004). The best configuration was determined to be the portfolio with the highest Sharpe Ratio (Sharpe 1966). By calculating Value at Risk (VaR) at both 95

Analysis methods centred around the use of statistical techniques. In order to normalise price data and enable meaningful asset-to-asset aggregation, log returns were calculated for every stock (Ruppert 2010). To estimate volatility and expected returns, summary statistics including mean and standard deviation were computed. Scatter plots of the efficient frontier were used to illustrate the results of Monte Carlo simulations, with the portfolio with the highest Sharpe Ratio prominently displayed. VaR measures, which provide measurable risk evaluations, were developed using the standard deviation and expected return of the portfolio. After combining all of these components, the finished application was uploaded to Streamlit Cloud, allowing for real-time, user-driven analysis via an interactive web dashboard.

0.5 System Design and Architecture

The AI-powered stock portfolio optimiser’s system architecture was created with cloud deployment in mind to guarantee scalability, responsiveness, and accessibility. Data storage, computation, risk analysis, portfolio simulation, and visualisation are all seamlessly integrated into the architecture. The system’s main parts are the local storage of historical stock data (CSV files), the use of Python-based libraries for preprocessing and analysis, modules for simulation and optimisation, and cloud deployment via Streamlit Cloud for interactive access ([Streamlit Inc. 2022](#)).

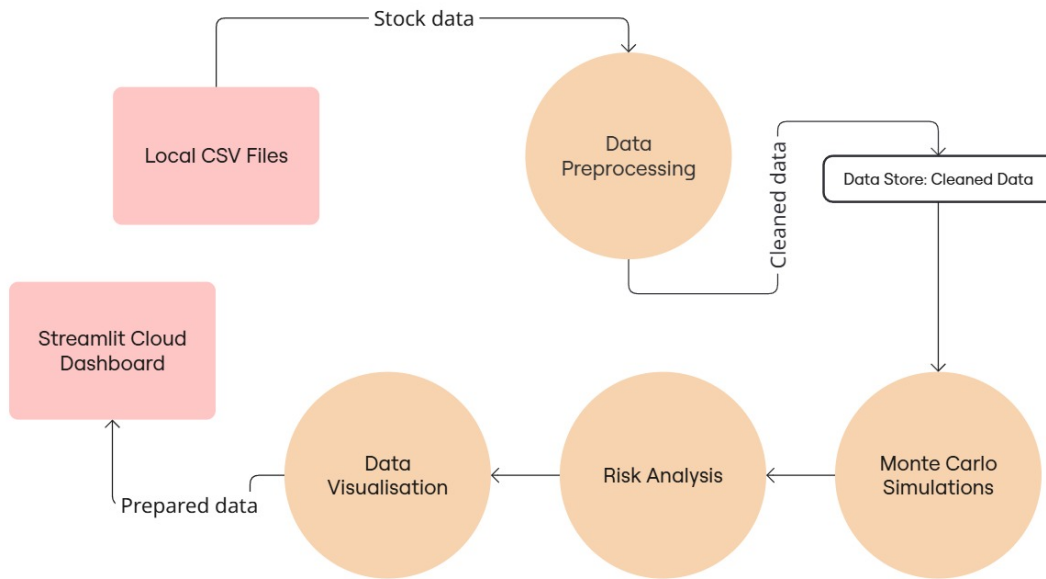


Figure 1: Data Processing Diagram for this project.

Initially, the raw stock trading data was stored in CSV format within a structured local directory. Upon launching the application, the data is ingested using Pandas, where it undergoes preprocessing including merging date and time fields, creating a consistent datetime index, and filtering for the last five years of data ([McKinney 2012](#)). The clean data is then fed into the simulation engine powered by NumPy and SciPy, where thousands of random portfolio weightings are generated ([Virtanen et al. 2020](#)). Each portfolio’s expected return and volatility are calculated, and the Sharpe Ratio is used to determine the optimal allocation ([Sharpe 1966](#)). Concurrently, the Value at Risk (VaR) metrics are calculated to assess potential portfolio losses at varying confidence intervals ([Jorion 2007](#)).

The results of the simulations, including the efficient frontier plot, optimal portfolio identification, and VaR metrics, are dynamically rendered through Streamlit components. The application is deployed on Streamlit Cloud, which handles the hosting and scaling of the dashboard without requiring manual server management ([Streamlit Inc. 2022](#)). Version control is maintained through GitHub, ensuring the system’s reproducibility and facilitating continuous updates ([Chacon & Straub 2022](#)).

Scalability was one of the main design factors for this architecture since Streamlit Cloud pro-

vides flexible resource management that can adjust to user demand. Another consideration was cost-effectiveness; the free tier of Streamlit Cloud was adequate for the scope of the project, preventing unnecessary spending ([Amazon Web Services 2024](#)). During development, security was controlled by keeping private repositories and making sure that no private or sensitive information was processed by the system. Additionally, simplicity and modularity were given top priority, guaranteeing that any part could be independently expanded, tested, and maintained without interfering with the pipeline as a whole.

0.6 Implementation

The cleansing and processing of the high-frequency trading data was the first step in the multi-phase development process for the AI-powered stock portfolio optimiser. The Date and Time data were combined into a single Datetime index by importing and processing the raw CSV files using the Pandas library ([McKinney 2012](#)). This allowed for consistent and efficient time-series operations. To make sure the study was still applicable to the current condition of the market, the datasets were then narrowed down to just include trading data from the previous five years. Dealing with inconsistent timestamp formatting and missing information across several stocks presented a significant issue during this phase. To fix these problems and maintain the integrity of the time-series alignment, data validation procedures were used, and any incomplete records were removed ([Tsay 2010](#)).

Following successful data preparation, the simulation engine was implemented. The system produced thousands of random portfolio weightings using NumPy ([Harris et al. 2020](#)). The Sharpe Ratio was employed as the performance criteria to choose the best portfolio after the expected return and volatility of each portfolio were calculated ([Sharpe 1966](#)). Normalisation following random generation was necessary to guarantee that portfolio weights always added up to one in order to incorporate Monte Carlo simulations ([Glasserman 2004](#)). Using SciPy, the risk analysis component was created ([Virtanen et al. 2020](#)). It provides users with information about the maximum possible losses over a certain time horizon by computing Value at Risk (VaR) for the optimised portfolio at both 95% and 99% confidence levels ([Jorion 2007](#)).

Within a GitHub repository, the project's code was arranged in a modular fashion ([Chacon & Straub 2022](#)). Reusable Python scripts can be found in the /src folder. These include modules for risk calculation (risk.py), data analysis (analysis.py), simulation (simulation.py), and data loading (load_data.py). The primary Streamlit application is the stock_portfolio_app.py script, which unifies all features into a unified dashboard. To facilitate environment setup, a requirements.txt file was included, which featured a list of all required Python dependencies. The repository was prepared for collaboration, backup, and incremental development by adhering to version control best practices.

Streamlit Cloud was used for cloud deployment, providing a simple and effective way to make the application available everywhere ([Streamlit Inc. 2022](#)). GitHub was used to link the project repository to Streamlit Cloud, enabling automatic deployment whenever new updates were released. The main application file (stock_portfolio_app.py) had to be specified, and the requirements.txt file had to contain a list of all required libraries. After the initial setup, Streamlit Cloud required little manual intervention to handle environment creation, dependency installa-

tion, and server administration.

The results of the implementation were highly successful. Large quantities of minute-level stock data were successfully processed by the program without experiencing appreciable performance decrease. In real-time, the Monte Carlo simulation engine produced thousands of possible portfolio combinations, and the optimiser quickly found portfolios with the best Sharpe Ratios. Users could modify simulation parameters and see fresh results thanks to the interactive dashboard and the helpful VaR outputs from the risk analysis component. All things considered, the system achieved all performance, functionality, and cloud deployment goals, resulting in a scalable, easily navigable prototype for portfolio optimisation.

0.7 Data Analysis and Results

The project's data preprocessing stage was essential to guaranteeing the precision and dependability of the ensuing simulations and analyses. Initially, CSV files for four significant tech companies (Apple, Amazon, Google, and Tesla) were used to import raw minute-level trading data. To facilitate consistent time-series analysis, the distinct Date and Time columns were combined into a single Datetime index using the Pandas package (McKinney 2012). In order to preserve current market relevance, the datasets were then filtered to only contain trading data from the previous five years. To avoid biases in the analysis, any rows with missing or invalid values were eliminated in addition to filtering. In order to normalise percentage fluctuations and enable meaningful aggregation across various assets, the price data was converted into log returns (Ruppert 2010).

Monte Carlo simulation methods were used for the analysis. NumPy was used to generate thousands of random portfolio weight combinations, making sure that the total of all the weights added up to one (Harris et al. 2020). The asset log returns weighted by the corresponding portfolio allocations were applied to each simulated portfolio to determine the projected return. The covariance matrix of returns, which took into consideration the correlations between various assets, was used to compute portfolio volatility (Glasserman 2004). To determine which portfolio offered the best risk-adjusted return, the Sharpe Ratio was then calculated for each one (Sharpe 1966). Value at Risk (VaR), which provides information on the maximum projected portfolio losses at both 95% and 99% confidence levels for a specified holding time, was computed utilising statistical techniques from SciPy in addition to optimisation analysis (Jorion 2007, Virtanen et al. 2020).

Using Matplotlib, the analysis's findings were displayed (Hunter 2007). The efficient frontier was visually represented by creating a scatter plot of all the simulated portfolios, with volatility on the x-axis and expected return on the y-axis. To differentiate the ideal portfolio from the larger cloud of portfolios, the one with the highest Sharpe Ratio was marked with a star. A clear visual breakdown of the suggested investment proportions across Apple, Amazon, Google, and Tesla was provided by the bar charts that showed the asset weight distribution for the ideal portfolio. On the Streamlit dashboard, VaR results were also shown quantitatively and updated dynamically according to user-selected criteria.

The interpretation of the findings indicated that the optimiser was effective in balancing risk and return across the available assets. Portfolios located towards the upper-left region of the ef-

efficient frontier exhibited the most favourable trade-offs between expected return and volatility (Elton et al. 2011). The Sharpe Ratio maximisation technique successfully identified portfolio allocations that achieved superior risk-adjusted returns compared to randomly weighted portfolios. The Value at Risk analysis reinforced the risk assessment by quantifying the potential losses users could face under normal market conditions. Overall, the findings demonstrated that a systematic, data-driven approach using historical stock data, combined with modern simulation and cloud technologies, can significantly enhance the process of portfolio optimisation.

0.8 Discussion

The main conclusions of this study show that leveraging contemporary Data Science, Big Data, and cloud computing platforms to create a cloud-deployed, AI-powered stock portfolio optimiser is completely possible (McAfee & Brynjolfsson 2012, Armbrust et al. 2010). Through the implementation of Monte Carlo simulations, Value at Risk (VaR) computations, and Modern Portfolio Theory (MPT) on high-frequency stock data, the team successfully developed a system that efficiently balances risk and return. Users are able to modify simulation parameters, examine thousands of possible portfolios, and comprehend the risk exposure of their optimal investments thanks to the interactive dashboard (Streamlit Inc. 2022). These results immediately address the initial study problem, which was to provide a data-driven portfolio management tool that was intelligent, scalable, and accessible while removing many of the subjectivities and inefficiencies that come with manual techniques (Fabozzi et al. 2007).

When compared to existing research and tools, this project offers several enhancements. Traditional financial modelling often remains either highly theoretical or confined to complex, institutional-grade software (Demyanyk & Kolliner 2018). Tools like Excel-based optimisers or specialised financial platforms such as Bloomberg Terminals are powerful but inaccessible to the general public due to their complexity or high costs (Bloomberg L.P. 2024). In contrast, this project builds on foundational financial theories while making the optimiser freely accessible through a cloud-deployed Streamlit dashboard. Moreover, while other studies have demonstrated Monte Carlo simulations or VaR calculations independently, few have combined them into an integrated, educational tool that users can interact with in real-time through a simple web interface.

These findings have important implications for the fields of cloud computing and big data. This project serves as an example of how cloud platforms may now provide robust, data-intensive financial analysis without the need for costly proprietary software or substantial on-premises equipment (Google Cloud 2024). The integration of interactive visualisation, scalable cloud deployment, and high-volume data handling provides a workable example of how Big Data initiatives might be made more widely available. Through advancements in cloud technology, tools that were previously only accessible to huge institutions are now accessible to individual users, reflecting a larger trend towards democratising advanced analytics (Hashem et al. 2015).

However, a number of restrictions were faced during the process. Firstly, the optimiser's predictive ability is limited by its reliance on historical stock data; past performance does not ensure future results (Bodie et al. 2014). The range of possible portfolios was constrained by the datasets' restriction to four significant technology equities. Furthermore, Monte Carlo

simulations are based on assumptions of normality and stationarity, which are not necessarily true in actual financial markets, even if they offer reliable testing in a variety of circumstances (Glasserman 2004). Additionally, real-time data feeds and predictive machine learning models, which could improve the optimiser’s responsiveness to market conditions—were not included in the project. Lastly, even though Streamlit Cloud managed the deployment effectively at this scale, future larger datasets or more complex simulations may call for more potent cloud infrastructure.

By incorporating real-time data streams via APIs like Yahoo Finance (Yahoo Finance 2024) or Alpha Vantage (Alpha Vantage Inc. 2024), further research could expand on this idea and enable dynamic portfolio rebalancing based on current market conditions. Portfolio diversity would be improved by extending the asset universe beyond four technology businesses to encompass other industries and regions. Additionally, the optimiser may be able to anticipate changes in the market and make more proactive adjustments to allocation techniques by integrating reinforcement learning or other machine learning algorithms (Li 2019). Lastly, putting the system on more scalable cloud platforms, such as Google Cloud Compute Engine or AWS EC2, may enable even bigger datasets and more computing demands, guaranteeing that the solution will continue to work as the system expands. To support scalability and real-time data processing in the future, I plan to integrate Apache Spark into the system architecture. Apache Spark offers robust distributed computing capabilities, making it particularly well-suited for handling high-volume financial data, especially when expanding the asset universe or incorporating live data streams from APIs. By using PySpark, the optimiser could efficiently process streaming data for dynamic portfolio rebalancing, perform large-scale Monte Carlo simulations in parallel, and compute risk metrics such as Value at Risk (VaR) across massive datasets with reduced latency. The integration would allow the system to evolve into a real-time decision support tool, capable of reacting to shifting market conditions and supporting more complex financial models across distributed environments.

0.9 Conclusion

This project set out to develop an AI-powered stock portfolio optimiser that integrates data science methodologies, big data handling, and cloud deployment to solve the problem of inefficient, manual investment decision-making. Through the use of Modern Portfolio Theory (MPT) ([Markowitz 1952](#)), Monte Carlo simulations ([Glasserman 2004](#)), and Value at Risk (VaR) calculations ([Jorion 2007](#)), the system was designed to analyse high-frequency trading data and optimise stock allocations for better risk-adjusted returns. The methodology involved careful preprocessing of minute-level stock data, statistical modelling of portfolio outcomes, and the development of an interactive Streamlit dashboard deployed via Streamlit Cloud ([Streamlit Inc. 2022](#)).

The findings demonstrated that it is possible to deliver a scalable, accessible, and educational financial analysis tool capable of assisting users in constructing optimised portfolios with a clear understanding of associated risks. The approach used highlighted the advantages of combining classical finance theories with modern computational techniques and cloud-based accessibility.

The contributions of this project are notable both technically and practically. Technically, it combined multiple financial theories and big data techniques into a single, cohesive system that can be used without specialised hardware or software. By deploying the optimiser to the cloud, the project democratises access to advanced financial analytics, offering a publicly accessible tool that individuals and small organisations can utilise ([Hashem et al. 2015](#)). From a methodological perspective, the integration of Monte Carlo simulations and VaR risk assessments within a real-time, user-driven application adds practical value, differentiating it from purely theoretical or isolated implementations in existing literature.

In conclusion, this project highlights the transformative potential of combining Big Data and Cloud Computing in building intelligent, accessible tools for financial decision support. It demonstrates that with the right integration of technologies, advanced portfolio management techniques can be made available to a much broader audience beyond institutional investors. The success of this project suggests promising opportunities for further expansion, innovation, and application of cloud-based, data-driven solutions in finance and other industries moving forward ([McAfee & Brynjolfsson 2012](#), [Armbrust et al. 2010](#)).

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